

EMPLOYMENT PASSPORT

*A Corporate Transparency Platform for
Bangladesh's Labour Market*

Category: Blockchain (Student)

Primary Domain: Identity & Privacy · Secondary: E-Governance · Tertiary: Fintech

TEAM 521.INC

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Platform

Hyperledger Fabric (permissioned), with daily Merkle-root
anchoring to a public chain

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1. Executive Summary

Bangladesh moves more than a million workers abroad each year and receives about USD 30 billion in remittances, yet the systems that carry those workers — and the companies that employ them — run on paperwork that is easy to forge and impossible for a stranger to verify. The same failure repeats at every level: a worker cannot prove her skill or wage history to a foreign employer; a foreign employer cannot tell a certified electrician from a forged certificate; a company cannot prove that it pays its workers or who actually owns it; an anonymous online review of an employer, however sincere, carries no more weight than gossip; and a rural jobseeker has no way to tell an honest recruiting agency from a fraudulent one before paying it.

Employment Passport solves this by becoming a corporate-transparency platform. Every fact about a person, a workplace, and a company — a skill, a job, a wage, a contract, an ownership stake, a review — is issued as a verifiable digital credential, signed by whoever is responsible for it, and anchored to a shared ledger whose history no single party can rewrite. From those credentials the platform assembles five surfaces:

1. **Professional identity** — a profile in which every skill and job is attested, never self-typed.
2. **Workplace transparency** — an employer review in which each review is provably written by a real employee, yet the author stays anonymous.
3. **Corporate identity** — a verifiable company registry, beneficial-ownership proof, and reconciled wage and procurement disclosure.
4. **Corporate Operations Suite** — a paid service that runs a workforce's lifecycle (onboarding, payroll, compliance, deployment logistics) so that managing workers is the same act as producing verifiable records.
5. **Labour-law enforcement** — a loop in which a worker's review reporting a violation automatically and provably reaches the ministry.

These five surfaces meet in a single transaction: the **job marketplace** (\$3.8), where an employer sees an application split into what is cryptographically proven and what an agency has merely asserted, and where an agency's standing rises or falls on the accuracy of the second category. This is where credentials stop being records and start being leverage.

The reason this must be a blockchain, and not another database, is the central fact of the environment: **institutions and paperwork in Bangladesh are not trusted, because they have repeatedly failed to be trustworthy**. A conventional database has an owner, and whoever owns it can change it — a corrupt office, a dishonest employer, or even the platform itself. Employment Passport removes that single point of trust: records are written only when the required parties co-sign them (an employer and a bank together sign a wage event), and the whole history is anchored daily to a public chain, so that not even the consortium behind it can rewrite the past without being caught.

The promise is not “trust us,” but “you do not have to trust anyone — verify the proof.”

For a worker, that means a credential that finally travels with her, and an agency whose honesty she can check before she pays. For a foreign employer, it means hiring a Bangladeshi worker without trusting any Bangladeshi institution. For a company, it means a verified reputation that wins government tenders and foreign contracts. For the government, it means a labour market whose violations surface automatically. And for everyone, it means a transparency platform that is itself incapable of quietly inflating or suppressing a score.

1.1 What we are asking the jury to evaluate

- **The problem is multi-party and trust-broken** — five parties (worker, domestic employer and bank, foreign employer, regulator and public, and the platform itself) whose incentives and mutual distrust no conventional system resolves.
- **Blockchain is necessary, not decorative** — §2.4 states precisely what a conventional database cannot do here, and §6.2 encodes that argument directly into per-schema endorsement policy.
- **Privacy is not sacrificed to transparency** — selective disclosure, zero-knowledge predicates, and a nullifier registry keep worker and company data out of the record while making the claims verifiable.
- **Incentives are mechanised, not asserted** — §3.8 shows the exact transaction in which an intermediary's honesty becomes cheaper than its dishonesty, and the safeguards that stop that mechanism being weaponised.
- **The architecture is specific, governed, sized and commercially real** — Hyperledger Fabric, six channels, per-schema endorsement policies, a nullifier registry, a specified anchoring bridge, and a capacity model benchmarked against published Fabric measurements (§6.10); funded by a paid Operations Suite priced against the domestic HRIS market, with a fee sanity-checked against current verification costs, a stated break-even, and a quantified Phase 0–1 funding gap.

2. The Problem

2.1 Scale and stakes

Bangladesh sent **1,130,757 workers abroad in 2025**, roughly two-thirds of them to Saudi Arabia alone, and the sector delivered remittance inflows of about **USD 29.6 billion** in the first eleven months of that year. The composition of that outflow is the crux of the problem: official manpower data for 2025 classified **43.47% of departing workers as less-skilled and 34.46% as semi-skilled**, with only 19.13% skilled and 2.94% professional. More than three-quarters of the country's most valuable export leaves without any portable, verifiable proof of what it can do.

Because skill cannot be proven, it cannot be priced. Bangladesh therefore competes on cost rather than capability, and workers absorb the difference. Official statistics put the average cost of overseas employment at roughly **BDT 4.17 lakh per migrant**, with male migrants paying around BDT 4.71 lakh and requiring an average of **17.6 months of foreign earnings simply to break even**. World Bank and KNOMAD survey work has repeatedly identified Bangladeshi workers as paying the highest worker-borne recruitment costs in the world. Earlier IOM survey work found that the large majority of what a migrant pays never reaches a licensed agency at all — it disperses among sub-agents and intermediaries whose only product is an introduction that cannot be verified.

Meanwhile, **demand-side verification is already arriving whether Bangladesh is ready or not**. A destination-country skill-verification programme now governs a large share of Saudi-bound migration; its capacity has scaled from roughly a thousand workers a month to about sixty thousand across twenty-eight approved centres, and it is now mandatory for visa issuance across thirty-three sectors (§5.1).

If Bangladesh does not build its own interoperable layer, its workers will be verified on someone else's terms, at someone else's price, with the data held by someone else.

The same trust deficit operates domestically. Roughly **20,000 computer-science and engineering graduates** enter the job market each year against only **7,500–8,000 IT openings**, forcing graduates to choose employers on marketing rather than evidence. With on the order of **2,400 licensed recruiting agencies** plus an informal sub-agent layer, a prospective migrant in a rural upazila has no way to check whether an agency is licensed or what it charged previous clients.

Across both markets the same three assets are at stake: a worker's history, a company's identity, and a company's reputation as an employer. All three exist today only as paper or as self-asserted claims — and paper is forgeable, while self-assertion is free.

2.2 Six concrete failure modes

(a) Credential fraud. Skill certificates, experience letters and trade-test results are printed documents with no machine-checkable authenticity. A forged five-year welding certificate is cheap; verifying a real one means phoning a training centre that may not answer, in a language the employer may not speak, in another time zone.

(b) Contract substitution and wage theft. A worker signs one contract in Dhaka and is handed a worse one on arrival. The original was never independently witnessed or timestamped, so the worker has no admissible evidence. Unpaid wages follow the same pattern: the only payroll record is held by the party accused of withholding.

(c) Recruitment opacity. A prospective migrant in a rural upazila cannot check whether an agency is licensed, what it charged previous clients, or whether the job order it advertises exists. Fraudulent agencies are indistinguishable from honest ones at the point of sale — and because the worker cannot tell them apart, the honest agency gains nothing from being honest. Opacity is not a side effect of the market; it is the market's operating condition.

(d) Unverifiable reviews. An anonymous employer review — however genuine — cannot be distinguished from a competitor's smear or a company's own astroturfing, so the market rationally discounts it to zero.

(e) **Labour law cannot be enforced.** Wage theft, unpaid overtime and unsafe conditions are rarely proven, because payroll and record both live in the employer's own database, which the investigator must take on faith.

(f) **Corporate opacity and career amnesia.** Who actually owns a company, and whether its disclosed wage bill matches what it pays, are unknowable without an audit the government cannot routinely perform. Symmetrically, a returning migrant's decade of experience is invisible to domestic employers, and a garment worker moving between factories loses tenure, safety training and machine-competency records at every transition. Experience the economy paid to create is destroyed on every move.

Note that failure mode (c) is not solved by better credentials alone. A worker with a perfect wallet still has to choose an intermediary, and the credential layer says nothing about whether that intermediary is honest. §3.8 addresses this directly.

2.3 Stakeholders and the five-party trust map

Each stakeholder needs something it cannot obtain alone, and — crucially — none of them trusts any other enough to build it.

Stakeholder	What they need	Why they cannot fix it alone
Worker / migrant	Portable proof of skill, contract terms, wage history; a way to tell honest agencies from fraudulent ones	Holds no institutional power; cannot compel any issuer to attest, and a self-asserted record is worthless
Recruiting agency / domestic employer	Cheap screening; credible differentiation from bad actors	Opacity is currently profitable for the worst actors, so the industry cannot self-regulate from inside
Destination / foreign employer	Confidence that a hire can do the job	Cannot audit foreign training centres or documents; defaults to discounting all claims, which depresses wages for everyone
Government (MoEWOE/BMET, RJSC, DIFE)	Reduced fraud, enforceable labour law, verifiable companies	A national database is not treated as authoritative by foreign employers, cannot bind foreign parties, and its own offices are not universally trusted
Bank / MFI / insurer; training institute (TTC)	Verified income for credit and insurance; recognition of certificates abroad	No trustworthy income signal for the informally documented, excluding millions from formal finance; a TTC is individually invisible and its brand cannot travel
The platform itself	Must be prevented from becoming the new single point of failure	Even the operator cannot be trusted to keep a neutral score when a paying client is involved

The pattern is the point: the problem is not that the data does not exist — it is that **no party is trusted to hold it, and every party's incentive is to manipulate it**. This is a coordination-and-trust failure, not an information-storage failure — precisely the class of problem a shared, governed, publicly-anchored ledger is built for.

2.4 Why a conventional database does not solve this

A conventional database — PostgreSQL, or any central system — has exactly one disqualifying property: **it has an owner**. Whoever administers it can update any row, delete any record, and rewrite history, and no outsider can tell. In this environment that single fact is fatal.

What the platform needs	What a database gives	What the ledger gives instead
A record no single office can fabricate	The admin can insert any row	Multi-party endorsement — a wage event is valid only if employer <i>and</i> bank co-sign
A history no one can silently rewrite	The admin can UPDATE / DELETE	Append-only, hash-chained blocks — tampering is detectable, not prevented by trust
“Was this revoked before or after the verification?”	A mutable column, answerable only by the admin	An ordered, tamper-evident status list every verifier reads
A reputation score no one can buy or edit	A number the operator controls, and can sell	Derived state — computed by chaincode from anchored events, recomputable by any member
Auditability by outsiders who hold no account	A read endpoint whose honesty must be assumed	A daily Merkle-root anchor to a public chain — verifiable by anyone
Trust between institutions that distrust each other	Each runs its own copy and they disagree	Joint governance with no single controlling party
Records that outlive their institutions	Survives only as long as its operator does	Credentials remain verifiable after an issuer leaves the network (§7.1)

The deeper reason is specific to Bangladesh: **any system whose integrity depends on the honesty of one administrator inherits that administrator's corruption**. The platform is built on the opposite assumption — that any single official, employer, agency, or even the platform operator may be dishonest — and is therefore designed so that no one of them can alter the record alone, and even all of them together cannot alter it silently.

Blockchain does not make dishonest people honest. It makes dishonesty **permanent, attributable, and expensive**, and it lets a foreign employer, a worker, or a journalist verify a fact without ever asking an institution for permission or trusting its word.

3. The Solution: a corporate-transparency platform

3.1 Concept

Employment Passport assembles five surfaces on top of one identity, credential, and anchoring layer, and joins them in one transaction layer. Each surface answers a different question about the same underlying trust gap, and each is unfalsifiable in the same way: a claim is only as good as the *signature* behind it, and a history is only as good as the *anchor* that makes it tamper-evident.

Surface	Question it answers	Built from
Professional identity	“What can this person provably do?”	Skill, contract, and wage credentials
Workplace transparency	“What is it actually like to work here?”	Verified-anonymous employee reviews
Corporate identity	“Who is this company, and what does it owe?”	Registry, ownership, procurement, wage disclosure
Operations Suite	“How does a company run a compliant workforce?”	The paid workflow layer that produces the records
Labour-law enforcement	“Does a violation reach the authority that can act?”	Review signals routed to the ministry

Surface	Question it answers	Built from
Marketplace (transaction layer)	“Whose word is this, and what is it worth?”	All of the above, at the moment of hiring

The design principle throughout is that **no surface stores the sensitive data it verifies**: a review never stores the reviewer's identity, a disclosure never the cap table, a wage event never the bank account. The ledger holds salted hashes and proofs; documents live in encrypted off-chain vaults (§6.4).

3.2 Credential types

Credentials are issued with BBS+ signatures, which allow a holder to prove an arbitrary subset of attributes — and predicates over them — without breaking the signature or revealing the rest.

Credential	Issuer	What it proves
SkillCertificate	TTC / trade-test centre / destination assessor	A standardised skill, level, assessment date
EmploymentContract	employer + agency (2-of-2)	The agreed terms, signed at onboarding
EmigrationClearance	BMET	Legally cleared to work abroad; corridor, job-order reference
WageEvent	employer + bank	A specific, dated wage payment; amount band; paid-on-time flag
EmploymentProof	employer	“This DID worked here, from A to B”
WorkplaceReview	worker (ZKP-gated)	A verified employee's assessment
ServiceRecord	employer at separation	Dates, role, conduct, reason for exit
AgencyLicence	BMET	Licence status, validity, permitted corridors
CompanyRegistration	RJSC	Legal existence and standing
BeneficialOwnership	company + auditor + RJSC	Ownership thresholds, proven selectively
Endorsement / ProcurementAward	peer or manager (staked); procuring entity	A revocable claim of competence; a tender outcome and conflict-check result

3.3 Selective disclosure in practice

Three worked examples show what “verifiable without revealing” means.

A bank reviewing a migrant-worker loan asks: “*paid at least twelve consecutive months at BDT 25,000+ in the past eighteen?*” The worker returns a zero-knowledge proof that answers yes; the bank learns nothing about the employer, the exact salary, or the country.

This predicate spans *multiple* credentials, which single-credential BBS+ disclosure does not give you on its own, so the construction is stated explicitly. Every WageEvent issued to a worker commits to the same holder-bound **link secret** (§6.5). The production path is the simpler of two: the remitting bank issues a rolling WageHistorySummary credential each month, over which an ordinary single-credential BBS+ predicate suffices — this is the default, because it removes multi-credential proving from a low-power device entirely. Where no summary issuer is available, the wallet falls back to a *conjunctive* proof over N wage credentials, demonstrating in zero knowledge that each is validly signed by an authorised issuer pair, that all N are bound to the same link secret and thus the same holder, that their periods are consecutive and fall inside the window, and that each amount satisfies the threshold.

A foreign employer asks: “*certified electrician, level 3+, no unresolved conduct finding?*” The proof does not reveal age, marital status, home district, or former employers' names.

A government tender asks a bidder: “*no beneficial owner above 25% who is sanctioned or a politically exposed person?*” The bidder proves the threshold holds without disclosing the cap table — the government sees compliance, not the company's owners.

3.4 Professional identity (the attested-profile layer)

A worker's profile is not a page of typed claims; it is a wallet of issued credentials, signed by the institutions responsible for them. Three properties distinguish it from a conventional professional network:

- **Attested endorsements.** A manager's endorsement is a staked, revocable credential, not a click. A false endorsement costs the endorser standing, so endorsements carry weight.
- **Corroborated reputation.** A claim supported by payroll *and* an independent bank record outranks a self-claim — the platform reports the evidence basis, not a bare score.
- **Privacy-preserving matching.** An employer searches “level-3 electrician, no conduct findings” and receives ZKP matches; the platform never sees the raw résumés it is matching.

3.5 Workplace transparency (the verified-review layer)

Existing employer-review sites face a fatal weakness: **anyone can post, and no one can tell a real employee from an astroturfing campaign**. Employment Passport resolves this with **verifiable anonymity** — the reviewer is provably an employee, and provably anonymous, at the same time.

The mechanism is the **nullifier registry**. A reviewer proves, in zero knowledge, that she holds a valid, unrevoked EmploymentProof for the company she is reviewing, and submits a one-time value $n = \text{PRF}(\text{linkSecret}, \text{companyDID})$ that is unique to (this worker, this company) and reveals neither. Chaincode accepts the review only if n has never been used. The result is **one verified voice per employee per employer** — a company cannot stack its own ratings, and no single employee can post a hundred. Because each employer relationship uses a distinct link secret, a worker's reviews across different employers cannot be linked to each other or to her. The full construction, including how the link secret is bound at issuance, is in §6.5.

What the public sees is not the raw reviews but a **transparency digest**: “*12 verified reviewers · 74% recommend · no open conduct findings*” — each figure a commitment whose root sits in the anchored Merkle tree, so the aggregate can be verified without ever reading a single review.

3.6 Corporate identity and the Operations Suite

The corporate layer turns the company itself into a verifiable subject. Every firm holds a **company DID** backed by a regulator-verified CompanyRegistration; ownership is proven through beneficial-ownership threshold proofs rather than a published cap table; procurement awards are anchored with their conflict-check results; and a company's disclosed wage bill can be reconciled against the same anchored wage events that power the reviews.

The **Corporate Operations Suite** is the paid application that makes corporate participation a product rather than an obligation. It manages the workforce lifecycle — onboarding and bulk credential issuance, payroll with wage-event anchoring, clearance and visa tracking, deployment logistics, benefits, and offboarding — so that *running* a compliant workforce is the same act as *producing* the verifiable records that build trust.

The commercial logic is deliberately tight: a corporate customer pays because the Suite produces the **trust signal that foreign employers and the government both rely on**. But the Suite is an **unprivileged client** of the same chaincode under the same endorsement policies as everyone else — a customer buys the *tooling*, never the *score*.

3.7 Labour-law enforcement (the review to ministry loop)

Reviews are not only reputation; they are the **earliest available sensor for labour-law violations**. A worker knows she was not paid before any inspector does, and the platform routes that signal to the authority that can act.

- Each review carries a **violation taxonomy** — *wage paid on time? contract honoured? overtime paid? conditions safe? termination dues settled?* A “no” is a violation flag, not a sentiment.
- The flag is **verified and corroborated** before it escalates: the reviewer holds a valid employment proof, and the signal is either matched against anchored wage events or confirmed by independent reviews above a threshold.
- When the threshold is met, chaincode emits ViolationSignal(companyDID, code, evidenceHash) and the platform routes a **scoped alert** to the ministry (MoEWOE / DIFE) — with the evidence, **without the reviewer's identity and without any free text**.

The ministry investigates under its own legal authority and anchors an InspectionReport outcome, which updates the company's transparency score. The loop therefore closes: a worker reports, the ministry acts, and the outcome is itself verifiable. Two guardrails keep it legitimate — **the platform supplies evidence, it never adjudicates**, and **selective disclosure prevents surveillance** (the ministry receives only the facts the specific case requires, with legal authority or worker consent).

THE ENFORCEMENT LOOP CLOSSES — a worker reports, the ministry acts, the outcome is itself verifiable



Figure 6 — The review-to-ministry enforcement loop.

3.8 The job marketplace and agency accountability

A credential is only worth what it does in a transaction. The marketplace is where issuance, verification, and reputation meet — and it is where §2.2(c), recruitment opacity, is answered.

The flow. An employer posts a JobOrder to the recruitment-channel. A worker applies directly, or — for the large segment lacking the literacy, connectivity or confidence to self-serve — through a licensed agency. **Agency-mediated application is the default path for the semi-skilled corridor, and the design treats it as a channel to be made accountable rather than a problem to be removed.** The employer reviews applicants on one dashboard: the worker's presented credentials, and where an agency submitted, its licence status and standing digest. **Every application is a bundle of two distinct kinds of claim, and this distinction carries the entire incentive.**

The employer's dashboard shows both classes, visually separated. Attested fields carry a verification result; asserted fields carry the agency's signature and nothing else. **An employer can therefore see at a glance how much of an application is proven and how much is an agency's word — and price the difference.**

This split is what makes the mechanism work. A dishonest agency cannot forge an anchored credential; the ledger check fails and the application is rejected, with no reputational event because no deception was possible. The fraud that actually occurs lives entirely in the second column — the five years of experience nobody documented, the fluency nobody tested. **Staking the agency's standing on the second column is the only version of this mechanism that bites.**

The reputation mechanism generates the credential demand the network needs

Every claim an agency can move from the asserted column to the attested column is risk it no longer carries. Agencies therefore acquire a direct commercial interest in getting their workers properly credentialed — which is precisely the issuance demand that solves cold start (§5.3).

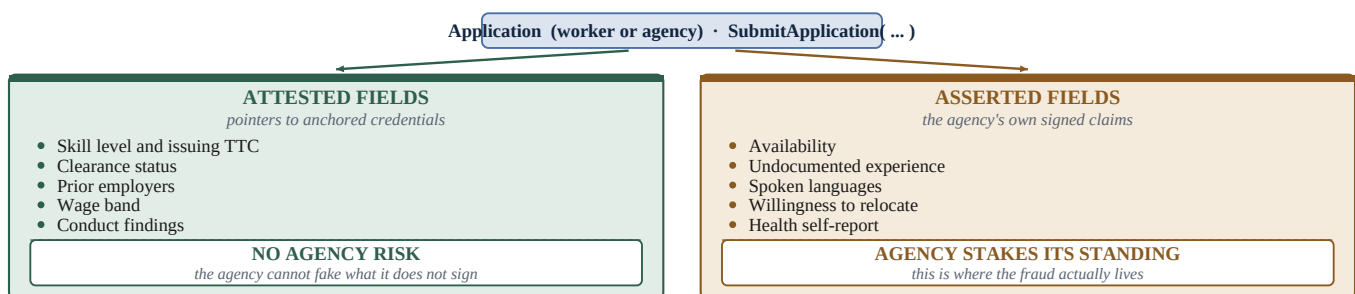
How standing changes — and how it cannot. An employer that discovers a mismatch does not adjust anything. It files an **allegation**, which anchors as an allegation and not as a finding. Three tiers:

1. **Automatic contradiction.** An asserted claim is directly contradicted by an anchored credential — the agency asserts no prior conduct finding where one exists. Chaincode flags this with the contradiction itself as evidence. No human judgement, no dispute, immediate effect.
2. **Post-placement non-corroboration.** The employer's ServiceRecord, anchored at separation, fails to corroborate what was asserted. This is the important tier, because most agency fraud is undetectable at screening and surfaces only after the worker arrives and the job does not match what was sold. The evidence arrives months late by design — and the agency cannot manufacture it, because it is signed by the employer.
3. **Contested allegation.** Everything else. The employer anchors the claim; the agency has a **14-day response window** in which to anchor a counter-claim; standing changes only after the window closes and UpdateAgencyStanding receives the **regulator + observer dual endorsement** it already requires (Appendix A).

No party adjusts another party's standing alone. An employer in a fee dispute with an agency, or one quietly owned by a competing agency, cannot damage it. This constraint is not a policy promise — it follows from the same endorsement architecture that governs every other consequential write (§6.2), and an allegation that never receives endorsement never becomes a finding.

Standing is computed, not tallied. A negative-only score would rank an agency with zero placements above one with a thousand, and would tell a worker to choose the emptiest agency — the opposite of the intent. Standing is therefore derived from four inputs: **placement volume**, the **corroboration rate** of asserted claims, **upheld disputes**, and **counterparty diversity**. The last defeats collusion: an agency and a friendly employer trading clean outcomes between themselves accumulate volume but not diversity, and score accordingly. Standing is published as a digest with its sample size, never as a bare number.

AN APPLICATION HAS TWO FIELD CLASSES — and only one carries risk



A MISMATCH CANNOT BE PUNISHED UNILATERALLY

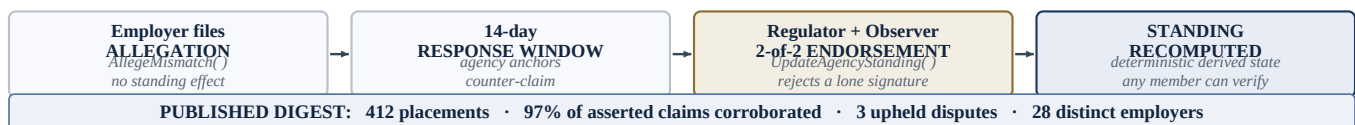


Figure 5 — The marketplace: attested versus asserted fields, and the governed path from an alleged mismatch to a standing adjustment.

The sub-agent question, answered plainly. The informal sub-agent layer is unlicensed, holds no DID, and will not be onboarded. We do not pretend otherwise. Standing binds to the **licensed agency**, which carries the full reputational cost of whatever its sub-agents assert in its name. This is the correct allocation: the agency chooses its sub-agents, profits from them, and is the only party with both the information and the leverage to discipline them. Making the agency liable for its own distribution network requires no new infrastructure and no fiction about registering thousands of village intermediaries.

Symmetry. The marketplace works in both directions: a worker considering a placement can retrieve the employer's transparency digest (§3.5) and the agency's standing digest before committing, by app, USSD, or one ID lookup at a bank branch or district office. §2.2(c) describes a prospective migrant with no way to distinguish a fraudulent agency from an honest one at the point of sale. **This is the answer: the question “is this agency honest?” acquires, for the first time, a checkable answer available before payment.**

3.9 End-to-end journey

A worker earns a level-3 electrician SkillCertificate at a TTC and is issued an EmigrationClearance by BMET. Unable to navigate a foreign application herself, she goes to a licensed agency — but first checks its standing digest by USSD and sees 400+ placements across 28 employers with a 97% corroboration rate. The agency submits her application against a Gulf employer's job order: her skill level and clearance are attested and verify automatically against the ledger; her claimed two years of informal workshop experience is an asserted field the agency signs for.

The employer verifies her credentials in seconds against the public anchor — never contacting a Bangladeshi institution — sees exactly which parts of the application are proven and which are the agency's word, and hires. An EmploymentContract is co-endorsed by employer and agency at onboarding. Her monthly WageEvent, co-signed by the bank, quietly builds a verifiable income history that later supports a loan predicate without revealing her salary.

Two years on, the employer anchors a ServiceRecord at separation. It corroborates the informal experience the agency asserted, so the agency's standing rises on evidence it could not have fabricated; had it not corroborated, the agency would have carried the cost, not the worker. If a subsequent employer fails to pay, she files a verified review flagging the wage violation; the corroborated signal reaches DIFE automatically, and the anchored InspectionReport records the outcome for every future worker. On return, her ServiceRecord makes a decade of experience legible to a domestic employer.

Across the whole journey, no single institution was trusted — and no single institution could have rewritten a single step.

4. Market and Partners

4.1 Market sizing and unit economics

We size the market on **verification and issuance events**, not on population, because revenue accrues per credential issued, per verification performed, and per worker managed.

Segment	Annual volume (order of magnitude)	Monetisable events	Indicative fee
Outbound migration	~1.0–1.1 M departures/yr	6–9 credential and verification events per worker per cycle: clearance, skill, contract, wage, service record, employer checks	BDT 100 / verification, tiered
Returning and repeat migrants	Several hundred thousand/yr	Re-verification for the next corridor; domestic re-entry	Highest willingness to pay — proven experience commands a wage premium
Domestic formal sector (RMG, manufacturing)	Workforce in the low millions, high inter-factory churn	Onboarding verification, safety-training proof, tenure	BDT 100 / verification
Financial services	Millions of remittance-receiving households	Income-verification API calls for credit, insurance, NRB products	BDT 25–50 / call
Regulator and agency tooling	~2,400 licensed agencies plus the TTC network	Seat SaaS and per-clearance licensing	BDT 50 / clearance licence
Marketplace	Job orders and applications across both corridors	Application screening; direct-hire success fees	Per-screening fee; negotiated success fee
Corporate Operations Suite	Employers and agencies running managed workforces	Tiered subscription by managed headcount + per-deployment logistics	BDT 150–200 / managed worker / month; BDT 1,000 / deployment

Unit economics. Marginal cost per verification is near zero: verification is an off-chain proof check plus a cached Merkle-path lookup (\$6.10), and it never touches consensus. Node capital and operating cost is distributed — each consortium member runs its own peer, so the foundation's cost base is engineering, common services, and support rather than infrastructure. We model foundation opex at roughly **BDT 50–60 M per year** (a ten-person engineering and support team plus common-service infrastructure).

Break-even, and why Phase 1 is deliberately sub-scale. At BDT 100 per monetisable event, break-even is approximately **550,000–600,000 events per year**, or about **70,000–75,000 fully enrolled workers** at eight events each — reached inside Phase 2 (\$8.3), before Suite or API revenue is counted. A 5,000-worker Phase 1 pilot generates only around BDT 5.5 M against that opex, so it is funded by grant, sponsor and regulator-integration contracts rather than transaction revenue; §8.3 quantifies the gap and names the sources. We state this plainly because a business model that pretends a pilot is self-funding is not one a jury should believe. The network expands corridor by corridor and crosses break-even at national clearance integration.

4.1.1 Is BDT 100 per verification realistic?

A per-event fee is only credible against what verification costs today. Four current reference points:

Verification pathway (current)	Cost	Payer	What it delivers
Educational / experience document attestation chain	BDT 3,000–4,000 per document	Worker or employer	One document, one jurisdiction, paper output
Destination-state skills verification test	USD 50 (≈ BDT 6,000); USD 50 per retake	Worker in practice	One trade, one corridor, five-year validity
Same test via intermediaries (actual worker outlay)	BDT 7,500–10,000	Worker	Same certificate, plus broker margin
Employment Passport verification	BDT 100	Verifier	Any anchored credential, any corridor, sub-second, reusable

At BDT 100, a verification is approximately **3% of the cost of attesting a single document** and **under 2% of the destination-state test fee**. The comparison is deliberately conservative: that programme *assesses* a skill while we *verify* an existing attestation, so the honest like-for-like comparator is the attestation chain, not the test. Even against that narrower benchmark, the reduction is roughly thirtyfold, and it is defensible on cost structure rather than on subsidy — a verification is an off-chain proof check plus a cached lookup, with marginal cost approaching zero, whereas an attestation is a manual, per-document, per-jurisdiction human process.

The risk is that BDT 100 is too low, not too high. A fee at 3% of the incumbent leaves substantial surplus unclaimed and moves break-even further out than necessary:

Fee per verification	Events needed for BDT 55 M opex	Equivalent enrolled workers (8 events each)	Still cheaper than attestation by
BDT 50	1,100,000	~138,000	98.5%
BDT 100 (base case)	550,000	~69,000	97%
BDT 250	220,000	~28,000	93%
BDT 500	110,000	~14,000	86%

At BDT 250 — still a 93% reduction against document attestation — break-even falls to roughly 28,000 enrolled workers, inside the Phase 1-to-2 transition rather than mid-Phase 2. We hold BDT 100 as the launch price for a deliberate reason: **the cold-start problem (§5.3) is worth more than early margin**. A verification cheap enough that a destination employer checks reflexively, rather than deliberating, is what drives the network-effect inflection in Phase 3. Pricing is a chaincode parameter under consortium vote (§4.3), so the network can move up the curve once verification is habitual, and the sensitivity table is the board's evidence base for doing so.

4.2 Partner ecosystem and their incentives

Partner	Role in the network	Their incentive to join
MoEWOE / BMET	Regulator node; issues Identity Anchor, Emigration Clearance and Agency Licence; enforcement responder; endorses standing adjustments	Fraud reduction, higher-skill export mix, a demonstrable worker-protection record, an automated violation early-warning system, real-time corridor analytics, and a working instrument for agency discipline
RJSC and BFIU	Company registry issuer; AML/CFT observer of disclosures	A registry actually trusted by banks, employers and foreign partners; threshold ownership proofs instead of full, sensitive cap tables
DIFE / labour courts	Enforcement verifiers and responders	Provable violations instead of testimony-versus-testimony
BAIRA + licensed agencies	Industry node; co-sign contracts; submit applications; operate the paid Suite	Faster placement, lower screening cost, and — for compliant members — a public standing digest that finally separates them from the actors described in §2.2(c)
Employers (domestic + destination)	Post job orders; issue employment proofs; consume verification; buy the Suite	Screening that distinguishes proven claims from asserted ones; a verifiable reputation that wins government tenders and foreign contracts
Banks / MFIs / Probashi Kallyan Bank	Co-sign wage events; issue income verification; node operators	A new creditworthy segment with verified income; lower default risk on migrant lending; remittance-product cross-sell
ILO, IOM, civil society	Observer nodes; audit and standards oversight; second endorsement on standing adjustments	Evidence base for fair-recruitment enforcement; independent verifiability of contract terms; a formal role in intermediary accountability
Foreign recruiters / employers	Verifiers; no network membership required	Self-contained proof with no local trust
TTCs, mobile operators and MFS providers, consortium foundation	Credential issuance; distribution, custodial recovery and USSD standing lookup; governance and network operations	TTC certificates become internationally checkable and agency demand for attested fields drives assessment volume; operators gain reach into the migrant segment; the foundation gains neutral, self-funding trust infrastructure

The government's five distinct roles are worth stating explicitly, because they are what make the network governable without making it capturable. It is a **regulator-issuer** (clearance, registry, agency licence), a **node operator**, an **oversight gatekeeper** (compliance veto, licensing, revocation, standing endorsement), a **verifier and consumer** (inspection, tribunals, public procurement), and a **policy setter** in the governance channel — but its sovereignty is bounded to its own acts, and even its history is publicly anchored. §7.1.2 addresses in full how state participation is bounded.

4.3 How partner incentives are allocated through the platform

Incentives are not promises in a partnership agreement; they are **encoded in the ledger and enforced by chaincode**, as network rights and revenue shares rather than as a speculative token.

- **Issuer revenue share.** Each paid verification is split by policy: a share to the credential issuer, a share to the network operations fund, a share to the worker-protection fund. Splits are chaincode parameters changeable only by consortium vote on the governance channel.
- **Reputation as an earned asset.** Agencies and employers accumulate an on-chain conduct record. High-standing members receive lower verification fees, priority in job-order matching, and an audit-lite compliance path. **Standing is derived state — computed by chaincode from anchored events, recomputable independently by any member — so it cannot be bought, and the platform operator cannot edit it.** §3.8 specifies the exact transaction in which it rises and falls.
- **Reputation stake.** Issuer standing rises on corroboration and is slashed on false attestation. Every issuer stakes its own reputation on every attestation it signs; every agency stakes its own on every asserted claim it submits.
- **Regulatory dividend.** BMET receives comprehensive analytics and audit capability, the automated violation early-warning loop, and a graduated instrument for agency discipline that sits below licence revocation — as public goods rather than paid lines.
- **Corporate trust signal.** A verified reputation is itself the incentive that makes corporate customers pay, because it converts into tenders and job orders won.
- **Worker-side cost of zero.** Workers never pay for issuance, storage, presentation, or for checking an agency's standing. The verifier pays. This is a deliberate design constraint: any worker-borne fee reproduces the extraction the system exists to eliminate.

5. Competition and Risks

5.1 Direct competition

Competitor class	Examples	Why we differ
Professional networks	LinkedIn	Attested, not self-typed: skills and jobs are signed credentials, and endorsements are staked
Employer-review sites	Glassdoor, Deshi Mula	Reviews are provably from real employees (verifiable anonymity), not anonymous-and-unverifiable
Job boards and recruitment portals	Bdjobs and equivalents	Carry self-asserted CVs and unrated intermediaries. No distinction between proven and claimed, and no consequence for an agency that misrepresents a candidate
Blockchain credential networks	Velocity Network Foundation, Dock	Built for literate, white-collar Western markets; assume employer-side HRIS integration. No migration clearance, contract substitution, wage, or enforcement coverage — the failure modes that dominate this corridor
Destination-state mandatory certification	Takamol Skill Verification Programme (Saudi Arabia); equivalent GCC schemes	Single-corridor, single-issuer, worker-paid, and non-portable. It certifies a <i>test result</i> ; it does not carry contract terms, wage history, conduct, or intermediary reputation, and its certificate is not recognised outside the issuing corridor. We integrate as an issuer and verifier rather than compete
Digital-locker schemes and generic SSI toolkits	Academic credential wallets; Indy/Aries, did:web	Locker schemes are education-only — employment, wage, conduct and ownership are out of scope. Toolkits give primitives, not a network: the hard part is not the cryptography but the consortium, the regulator mandate, and the corridor-specific schemas
HR / payroll SaaS	Workday; PiHR, SmartHRM, LinesPay, PeopleDesk (Bangladesh)	Their records live in the employer's own database, which is exactly the party a wage-theft investigator cannot take on faith. No multi-party endorsement, no external anchor, no worker-held copy

The Takamol case deserves specific treatment, because it is **simultaneously our most serious competitor and our strongest market validation**. Launched in Bangladesh in February 2023 under an MoU with BMET, the Skill Verification Programme has moved from optional to structurally mandatory: the Saudi embassy stopped approving visa applications without the certificate as of 19 June, and the requirement

expanded from five occupational sectors to thirty-three. Workers sit an online exam for a certificate valid five years, at a registration fee of USD 50; over 200,000 had been accredited as of October 2023. This proves the thesis of §2.1 in the sharpest possible terms: **destination-side verification is not hypothetical, it is already mandatory, and it is already monetised at roughly BDT 6,000 per worker** — on terms Bangladesh does not set. Of the fee, the administering TTC receives USD 13 and the remainder goes to the Saudi side; although the employer was expected to pay, the cost has in practice fallen on the worker, who pays BDT 7,500–10,000 through agents and computer shops. Bangladesh has formally asked Riyadh to reduce the USD 50 retake charge — the clearest available signal that the sending state has no leverage over pricing in a system it does not own.

Our position is complementary at the technical layer and competitive at the strategic layer. Technically, a Takamol result can be issued into the wallet as a SkillCertificate with Takamol as signing issuer. Strategically the programme is **single-corridor** (a Saudi certificate does not travel to Qatar, Malaysia, or a domestic employer), **single-fact** (a test result, not contract terms, wage history, conduct, or the intermediary's honesty), **institution-custodied** (the worker holds a PDF, not a key, and cannot present a selective proof), and **worker-paid**. Existing Bangladeshi certifications are not recognised, so workers re-certify at additional cost — the non-recognition problem an interoperable W3C credential layer solves, and the negotiating asset Bangladesh currently lacks.

On the domestic HRIS market, competition is well served and price-competitive — PiHR alone reports over 500 client companies — which is precisely why we do not compete on payroll features. A PiHR payroll record and an Employment Passport WageEvent describe the same underlying payment, but the first is a row in a vendor database that the employer can edit and that no foreign party will accept as evidence, while the second requires the bank's independent co-signature (§6.2) and is anchored beyond the employer's reach. **The barrier to entry we are building is not the software; it is the consortium.** Any competent local vendor could implement our chaincode interface. None of them can convene BMET, RJSC, BFIU, BAIRA, and the banks into a governed network with a compliance veto and no controlling member — and without that convening, the multi-party endorsement that makes the record worth anything is unavailable at any price. The same is true of the marketplace: a job board is easy, but a job board whose intermediary ratings cannot be bought requires a regulator willing to endorse standing adjustments. Our expected relationship with these vendors is integration, not displacement: a payroll adapter (§6.7) lets an employer keep its existing HRIS and anchor from it.

5.2 Indirect competition

- **Status quo paper and attestation chains** — notarisation, embassy attestation, physical verification visits, and the paid middlemen who “verify” them. Slow and expensive, but incumbent and legally familiar. **This is our real competitor**, and it is itself the vector for much of the fraud the platform removes.
- **Personal networks and village-level reputation** — today a worker's only real signal about an agency is whether a neighbour's placement worked out. This is genuinely informative and genuinely local; it does not scale, cannot be checked, and is exactly what a fraudulent agency exploits by moving districts.
- **Professional background-screening firms** — accurate but priced for corporate hiring, not for a construction employer screening two hundred labourers.
- **Social proof networks** — self-asserted and largely irrelevant to the semi-skilled segment.
- **Doing nothing and discounting** — employers simply assume all claims are false and pay accordingly. This is *rational today*, it costs the employer nothing, and it is exactly the equilibrium we intend to break. Any honest competitive analysis must name it as the default outcome if we fail.

5.3 Business risks and mitigations

Risk	Severity	Mitigation
Incumbent resistance — opacity is profitable for the worst intermediaries, who will lobby against adoption	High	Do not fight the middle layer — <i>re-price</i> it. Offer verified agencies faster clearance, a certified tier, and preferential job-order access. Anchor the mandate at the regulator and the destination-country demand side, where rent-seekers have least leverage
Two-sided cold start — credentials are worthless until verifiers check them; verifiers do not check until credentials exist	High	Seed the supply side with records already digital and already trusted (BMET clearance, TTC trade-test results). The marketplace field split (§3.8) gives agencies a commercial reason to drive issuance, since attested claims carry no risk. Launch one corridor with one large employer group. Launch pricing is set for reflexive adoption, not margin (§4.1.1)
Standing used as a weapon — an employer damages a rival's or a supplier's agency in a fee dispute	High	Employers file allegations, never findings. Standing changes only after the 14-day agency response window and regulator + observer dual endorsement (§3.8). An unendorsed allegation has no effect on standing
Collusive standing inflation — an agency and a friendly employer trade clean outcomes	Medium	Counterparty diversity is a scoring input, not a bonus. Volume without diversity does not raise standing
Sub-agent layer outside the system — the actual chooser is an unlicensed village intermediary with no DID	High	Standing binds to the licensed agency, which carries the full cost of what its sub-agents assert in its name. The agency chooses them, profits from them, and is the only party able to discipline them. No registration of informal intermediaries is required or attempted
Defamation or legal exposure from reviews and allegations	Medium	Credential-gated reviews, no free-text rendering in public digests, anonymous reporters, anchored outcomes. Allegations are visible as allegations, never as findings, until endorsed
Low digital literacy in the core segment	High	USSD and agent-assisted flows; custodial recovery; branch-based enrollment; Bengali-first wallet UX. Standing digests are retrievable by USSD or a single ID lookup. The worker's minimum required action is presenting a QR code or an ID number
Regulatory change, UBO disclosure resistance, partner non-cooperation on wage events, data-protection law	Medium	Pilot with one willing ministry and integrate as an <i>output</i> of existing processes; threshold ZKPs make compliance cheaper than concealment; dual-source wage events mean no party can veto the evidence; no PII on-chain, pairwise DIDs, and crypto-shredding erasure (§5.4)
Grant funding does not materialise	Medium	Invert the sequence: launch the Suite commercially on workflow value alone and fund consortium formation from revenue (§8.3)
Paying-client capture (score inflation)	High	The score is derived state, computed from anchored evidence and independently recomputable by any member. The platform <i>cannot</i> edit it — this is enforced architecturally, not by policy

5.4 Technical risks and mitigations

Risk	Mitigation
Key loss — a worker loses their phone and with it a decade of career history	Shamir 2-of-3 social recovery split across a BMET district office, a partner bank branch, and a family guardian; optional HSM-backed custodial mode with a documented path to self-custody
Garbage in, gospel out — a colluding employer issues a false credential	Corroboration scoring: a wage claim supported by both payroll and an independent bank credit scores higher than a single-issuer claim. Verifiers see the <i>evidence basis</i> , not just a boolean. Issuers stake reputation on every attestation
Review gaming via fresh link secrets	The link secret is committed inside the EmploymentProof at issuance; the nullifier proof demonstrates derivation from <i>that committed</i> secret (§6.5). A worker cannot mint a new secret to post a second review
Standing recomputation divergence	Standing is deterministic derived state computed by chaincode from anchored events. Any member can recompute it independently from its own ledger copy; divergence is a detectable consensus fault, not a silent error
Right-to-erasure versus ledger immutability	Personal data lives only off-chain under envelope encryption. Erasure is executed as crypto-shredding of the data key, leaving unreadable ciphertext and an unlinkable hash — satisfying erasure obligations without rewriting the ledger
Sybil issuers — fake employers or agencies onboarded	MSP identity only after regulator-verified legal existence; two-organisation onboarding endorsement; on-chain revocation. Agency membership additionally requires a valid AgencyLicence
Scalability under peak load; endorsement latency as the real constraint	Derived capacity model benchmarked against published Fabric measurements (§6.10): peak demand is under 3% of the most conservative reported throughput, and verification reads never reach consensus. Partner peer response times are bounded by network SLA (§7.2.3) and measured from Phase 0 against live bank adapters rather than synthetic peers
Correlation, ZK cost on low-end devices, crypto obsolescence, regulator overreach	Pairwise DIDs and per-employer link secrets defeat correlation; proofs verify off-chain at the edge and the WageHistorySummary path (§3.3) removes multi-credential proving from the device; the signature suite is a schema field so credentials are re-issuable under new suites; the platform holds no plaintext and cannot dump records on demand (§7.1.2)
Fabric single points (ordering, anchor bridge)	Raft ordering with one orderer per founding organisation; no organisation holds a majority; 4-of-6 threshold-signed anchor publication (§6.8); geographically distributed peers; documented RTO/RPO in the network SLA

6. Architecture

6.1 Platform selection

Employment Passport runs on **Hyperledger Fabric (permissioned)** with **daily Merkle-root anchoring to a public chain**. The choice is deliberate and matches the trust problem:

- **Permissioned membership** — every participant holds an MSP identity issued only after regulator-verified legal existence. This is what makes “who attested this” meaningful: a review, a wage event, an application, or a disclosure is signed by a known, licensed institution, not an anonymous public key.
- **Private data collections and channels** — sensitive operational data (contracts, review text, UBO cap tables, application detail) is partitioned so each organisation holds only what it is entitled to, while all hold the hash that proves integrity. This is the privacy model a public chain cannot offer natively.
- **Pluggable endorsement policies** — a write is valid only when the required parties co-sign it (employer + bank for a wage event; regulator + observer for a standing adjustment). This encodes “no single party can fabricate a record, and no single party can punish another” *directly into consensus*, which is the heart of the corruption argument.
- **Public anchoring** — the permissioned ledger alone still leaves “trust the consortium.” The daily Merkle-root bridge closes that gap: history becomes auditable by any third party holding a prior root.

The trade-off is stated openly: **Fabric gives governed multi-party trust, not permissionless decentralisation**. That is exactly what a regulated labour-and-corporate context requires, and the public anchor supplies the external accountability a public chain is good at.

System layers. Read top to bottom, each layer knows less about the worker than the one above it — the applications hold keys and plaintext, the services hold ciphertext and proofs, the ledger holds hashes and policy, and the public chain holds a single 32-byte number per day. That gradient is the architecture's central idea.

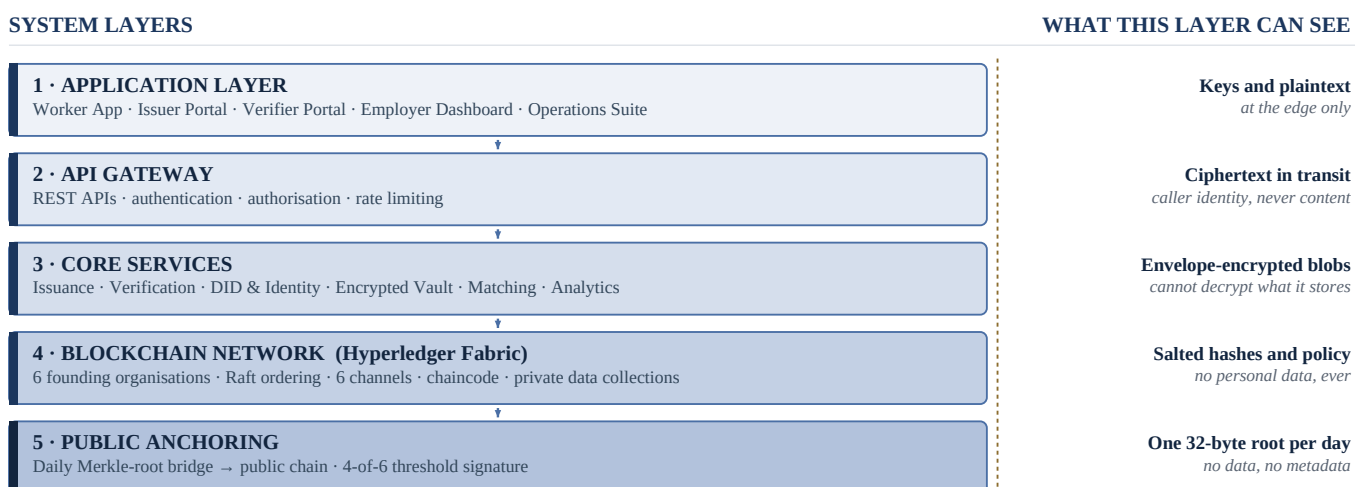


Figure 1 — The five-layer system architecture. Each descending layer holds strictly less about the worker than the layer above it.

6.2 Network topology and consensus

The founding consortium is six organisations, with **Raft** ordering and one orderer per organisation, so no single organisation controls ordering. Six channels partition visibility, and anything whose *mere existence* is sensitive lives in a private data collection within its channel.

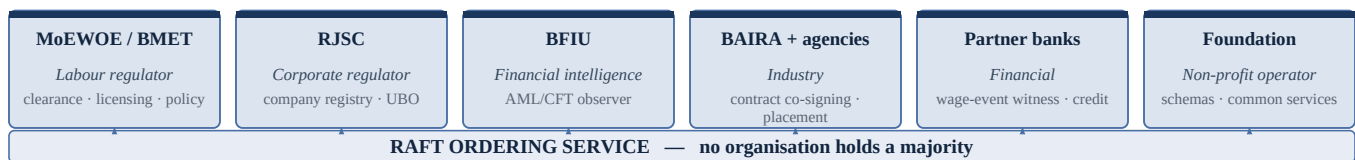
FOUNDING CONSORTIUM — Raft ordering, one orderer per organisation**CHANNELS — visibility partitioned; private data collections within**

Figure 2 — Founding consortium, ordering service, and channel segregation. The governance channel (gold) is restricted to founding organisations.

Endorsement policies are set per schema, so the quorum is a business decision, not a global constant. This table is where the corruption argument becomes executable code:

Schema	Endorsement policy	Rationale
EmploymentContract	employer + agency (2-of-2)	Neither party can unilaterally invent a contract
WageEvent	employer + bank	A wage record needs the payer <i>and</i> an independent financial witness
EmigrationClearance / AgencyLicence	BMET	Sovereign regulatory acts
CompanyRegistration / BeneficialOwnership	company + auditor + RJSC	Corporate identity is multi-party attested
WorkplaceReview	worker ZKP + nullifier check	Anonymity with one-voice guarantee (\$6.5)
Endorsement	endorser + subject consent	Attested, not self-typed
Application	submitter (worker or agency) + valid AgencyLicence where agency-submitted	The agency's signature is what binds it to its asserted claims
MismatchAllegation	employer alone	Anchors as an <i>allegation only</i> ; carries no standing effect
StandingAdjustment	regulator + observer (2-of-2), after response window	No party may damage another's standing alone. This is the safeguard that makes the reputation mechanism usable rather than weaponisable
ServiceRecord	employer, subject to a 30-day worker dispute window	Exit records are the most abuse-prone. A disputed record presents to verifiers as <i>contested</i> rather than as settled fact, and is resolved under §7.2.3

6.3 Channels and data segregation

Channel	Members	Contents
anchor-channel	All organisations	Credential hashes, DID document pointers, revocation status lists, schema registry
recruitment-channel	BMET, BAIRA, agencies, employers	Job orders, applications (attested references and asserted-claim hashes), clearance events, contract endorsements
payroll-channel	Banks, employers, BMET	Wage-event anchors, corroboration records
corporate-channel	RJSC, BFIU, BMET, banks, auditors, employers	Company registry, UBO threshold proofs, procurement awards, disclosures
reputation-channel	Employers, agencies, worker wallets, regulator, observer, foundation	Review anchors, nullifiers, endorsements, mismatch allegations, standing adjustments, aggregate digests
governance-channel	Founding organisations only	Policy votes, chaincode upgrades, membership changes, fee parameters, standing-formula parameters

Application detail — the full asserted-claim text, the worker's contact information, the employer's internal screening notes — lives in a private data collection on recruitment-channel visible only to the submitting agency, the receiving employer, and the regulator under legal authority. All organisations hold its hash and can verify integrity, which is what makes a later contradiction provable.

6.4 On-chain versus off-chain

The invariant is absolute and stated as a design rule: **no personal data is ever written to the ledger, in plaintext or in reversibly encrypted form**. The ledger holds what must be *provable*; the vault holds what must be *private*.

On-chain (ledger state)	Off-chain (encrypted vault)
Salted credential hash + schema identifier	Full credential JSON-LD document
Issuer DID + signature-suite identifier	Worker name, NID/passport, address, biometric references
Review hash + review-proof commitment + nullifier	Full review text, sentiment, free-text detail
Application hash + attested-credential references + asserted-claim hash	Full application, asserted-claim text, employer screening notes
Allegation hash + evidence hash + response status	Allegation narrative and supporting documents
Standing digest commitment (per agency, per company)	Individual rating and outcome breakdowns
UBO threshold proof (e.g. “no PEP ≥ 25%”)	Full cap table, nominee details

On-chain (ledger state)	Off-chain (encrypted vault)
Bitstring revocation status list; issuance and revocation timestamps; endorsement set and corroboration score; procurement award hash and conflict-check result; consent and disclosure audit hashes	Employer HR notes and any attribute subject to erasure obligations; contract PDFs, wage slips and assessment evidence; tender documents; media such as photographs, certificates and scanned documents

Off-chain storage is an IPFS-based private cluster replicated across member organisations, with per-credential envelope encryption; the data key is wrapped to the worker's DID key and to any keys the worker has explicitly granted.

6.5 Identity, credentials, and the nullifier registry

Identity stays **pairwise**: each worker and company uses a distinct DID per verifier relationship, so presentations cannot be correlated across verifiers. Credentials are issued with **BBS+ signatures**, preserving selective disclosure and predicate proofs over hidden attributes.

The **nullifier registry** is the “one voice” mechanism for the workplace-transparency surface. A review must be provably written by a *real* current or former employee, yet the reviewer must stay anonymous *and* unable to post many reviews for the same employer.

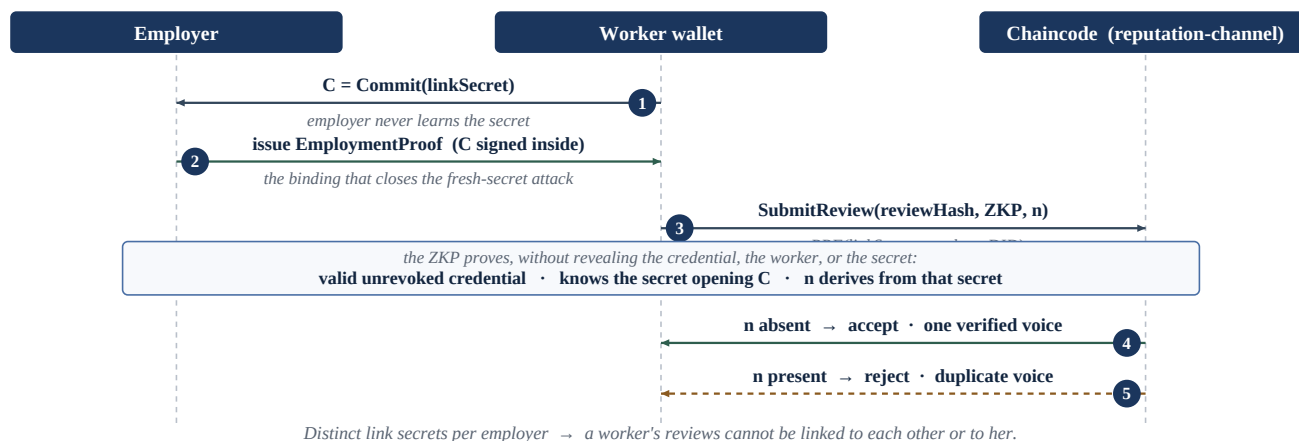


Figure 3 — The nullifier construction. Committing the link secret inside the signed credential at issuance is what closes the fresh-secret attack.

1. **At issuance**, the worker's wallet generates a per-employer linkSecret and sends its commitment $C = \text{Commit}(\text{linkSecret})$ to the employer. The employer issues an EmploymentProof credential that **includes C as a signed, blinded attribute**. The employer never learns linkSecret itself.
2. **To review**, the wallet derives a **nullifier** $n = \text{PRF}(\text{linkSecret}, \text{employerDID})$ — a one-way value unique to (this worker, this employer) and unrevealing of either.
3. The wallet submits a **zero-knowledge proof** that (a) it holds a valid, unrevoked EmploymentProof for employerDID, (b) it knows the linkSecret opening the committed attribute C inside that credential, and (c) n is correctly derived from *that* secret — without revealing the credential, the worker, or the secret.
4. Chaincode appends n to the nullifier registry **if and only if it is absent**. A second review with the same n is rejected.

Step 1 is what closes the obvious attack. If the link secret were merely chosen at review time, a worker could generate a fresh secret and post unlimited reviews. Binding the commitment inside the signed credential means there is exactly one valid secret per employment relationship, and therefore exactly one valid nullifier.

Result: one verified voice per employee per employer, with no linkage across a worker's different reviews (distinct link secrets per employer) and no identity exposure to the platform or the company. This is the cryptographic upgrade that turns anonymous gossip into audit-grade transparency. The same committed link secret is what enables the multi-credential wage predicate in §3.3: wage credentials issued to the same holder share the commitment, so the wallet can prove they belong to one person without revealing who.

Revocation uses a bitstring status list published per issuer, so a verifier learns whether *a* credential is revoked without revealing *which* credential it is checking. Schemas are version-controlled on the governance channel, so a credential's meaning cannot be silently changed after issuance.

6.6 Key management

Self-custody for smartphone users, with keys in device secure storage. **Social recovery** by Shamir 2-of-3 split across a BMET district office, a partner bank branch, and a family guardian. **Assisted custody** for feature-phone and low-literacy users through an HSM-backed custodian, with worker approval required per action, every action anchored, and a documented path to self-custody. **Organisational keys** live in HSMs with role-based quorum, so no single employee can sign an attestation, an application, or an allegation on behalf of an institution. Rotation is quarterly and mandatory (§7.1.4).

6.7 Integration with legacy systems

BMET clearance and agency licensing credentials are emitted as standard outputs of existing national processes — an *adapter*, not a replacement: the process the officer follows does not change. The **RJSC company registry** is *mirrored*, not rebuilt; RJSC remains authoritative and signs CompanyRegistration and UBO attestations while the ledger records the signed anchor. **Employer HR, payroll and ATS systems** integrate via a payroll adapter or scheduled file drop, including from existing domestic HRIS platforms, so an employer keeps its current vendor and gains anchoring; applications can be pushed into an existing applicant-tracking system with the attested and asserted split preserved as structured fields. **Banks, MFIs and insurers** send wage events by core-banking API or file drop, and the ledger receives the anchor, never account data. **Paper and USSD fallbacks** allow signed-QR verification and standing-digest lookup where no integration exists. **Foreign verifiers need no integration at all**: they check a self-contained BBS+ proof plus a public-chain Merkle proof — no Fabric membership, no API key, no “trust the ministry.”

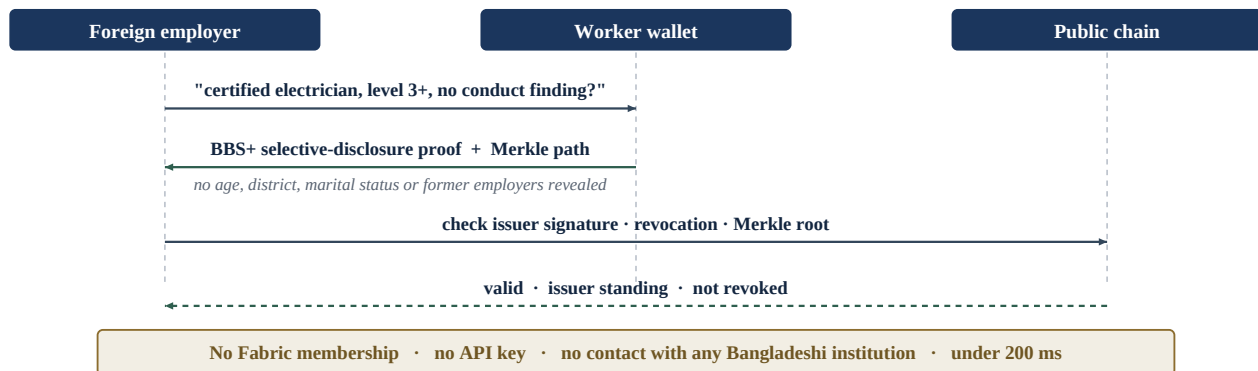


Figure 4 — Foreign verification without local trust. The verifier holds no account, signs no agreement, and contacts no Bangladeshi institution.

6.8 Public anchoring and the transparency digest

Once daily, the Merkle root of all anchors committed that day is written to a public chain. The published root is a single 32-byte value containing no data and no metadata about any individual. Its purpose is narrow and important: it makes the permissioned ledger's history **externally auditable**, so that even a hypothetical collusion of every consortium member could not rewrite past attestations without detection by any third party holding a prior root.

Bridge specification. The anchor target is a **public chain selected for settlement finality and fee predictability rather than ecosystem features**, since the payload is one 32-byte value per day and no smart-contract execution is required; Bitcoin via OP_RETURN and Ethereum calldata both qualify, and the choice is a governance-channel decision (§7.3.2), not an architectural dependency — the root is chain-agnostic and can be dual-published. The bridge is a **threshold-signed publisher, not a trusted oracle**: each founding organisation's peer computes the daily root independently from its own ledger copy, and publication requires a **4-of-6 threshold signature**, so no single member — including the foundation — can publish a root the others did not compute. A member computing a divergent root raises a governance-channel alarm before publication, making disagreement a detection mechanism rather than a failure.

Failure handling is deliberately undramatic, because the bridge is an audit convenience rather than a liveness dependency. A missed publication does not affect issuance, verification, or revocation, all of which run against Fabric state; missed roots are queued and published as a batch with original date stamps and inter-root hash links, so a gap is visible and reconstructible but not a loss of evidence. Only a sustained outage beyond thirty days escalates to a governance incident. **Fee settlement** runs through a regulated virtual-asset service provider under contract to the foundation: the foundation pays a fiat invoice, the provider publishes from its own address, and the provider's attestation is itself anchored. No consortium member holds, custodies, or is exposed to a digital asset at any point — a hard constraint given central-bank restrictions, and one that governs the choice of mechanism.

The **transparency digest** is the bridge's second, purpose-built output, published for both companies and agencies:

Company: 12 verified reviewers · 74% recommend · 3 wage events reconciled this month · UBO thresholds satisfied · no open conduct findings.
Agency: 412 placements · 97% of asserted claims corroborated · 3 upheld disputes · 28 distinct employers · licence valid.

Each figure is a commitment whose root is in the anchored Merkle tree, so the public can verify the aggregate is computed over the real evidence **without seeing any raw review, salary, owner, or application**. This is what makes the *platform itself* unable to quietly inflate or suppress a score — the corruption defence applied to the transparency layer, and the reason an agency cannot buy its way up the rankings.

6.9 The Operations Suite as an unprivileged client

The paid Operations Suite is an **application-layer client, not a privileged write path**. Its modules — onboarding and bulk issuance, payroll anchoring, clearance and visa tracking, deployment logistics, application submission, benefits, offboarding, analytics — all call the **same chaincode under the same endorsement policies** as any other participant. A corporate customer pays for *workflow, tooling and logistics*, not for a backdoor: the Suite cannot mint a wage event without the bank's endorsement, issue a clearance without BMET, delete a review, or **adjust its own standing or another party's**. An agency running placements through the Suite gets faster submission and better tooling; it does not get a better score. Running payroll and clearance through the Suite therefore *is* the transparency surface — the same act that manages the workforce produces the evidence the platform publishes.

6.10 Performance and capacity

Capacity is derived from the volumes in §4.1 rather than asserted.

Write source	Derivation	Writes per year
Outbound migration	1.13 M departures × 8 credential events	~9.0 M
Wage events (Phase 3 scale)	500 k enrolled workers × 12 months	~6.0 M
Domestic onboarding and service records	RMG and manufacturing churn	~1.0 M
Applications and job orders	~1.5 M applications at ~2 per placement	~1.5 M
Workplace reviews	500 k workers × ~20% review rate	~0.1 M
Allegations and standing adjustments	Exception path only	~0.02 M
Corporate (registration, UBO, procurement)	~2,400 agencies + registered employers	~0.05 M
Total		~17.7 M writes/yr

17.7 M writes over 31.5 M seconds is approximately **0.56 TPS average**.

Peak factor. Writes concentrate into business hours ($\approx 3\times$), working days ($\approx 1.4\times$), and month-end payroll batches plus seasonal clearance clustering ($\approx 5\times$) — a combined peak factor of roughly **20×**, giving **~11 TPS sustained peak**, with headroom modelled to 15 TPS.

Benchmark comparison. Fabric's headline figure is end-to-end throughput above 3,500 transactions per second with sub-second latency, scaling to over 100 peers, reported in the platform's reference paper.¹ We do not size against that number, for two reasons a technical reader would raise: it was measured on a UTXO-style chaincode rather than our credential-anchoring workload, and it predates Raft — subsequent benchmarking work notes that these results used Zookeeper or PBFT ordering and are not directly comparable to current deployments.²

We therefore size against the conservative end of the independent literature. Thakkar et al. report peak throughput of approximately 850 TPS,³ and a 48-node commodity-hardware evaluation reports 400 TPS.⁴ Recent Fabric v2.x measurement gives a peak of 1,821 TPS at two peers, declining to 993 TPS at sixteen peers as endorsement and validation load spreads across the network.⁵

Taking the most pessimistic of these — 400 TPS on commodity hardware — our 11 TPS peak is under 3% of demonstrated capacity, and our 15 TPS headroom case is under 4%.

The declining-with-peer-count trend in the v2.x results is the relevant one for us, since our six founding organisations each run a peer and our endorsement policies require two- and three-party signatures. It is also why we identify the true constraint as **endorsement round-trip latency to bank and regulator peers, not ordering throughput**: at 11 TPS the ordering service is idle, while a wage event awaiting a bank's endorsing peer, or a standing adjustment awaiting regulator and observer, is bounded by that partner's infrastructure and governed by SLA (§7.2.3) rather than by consensus. This is the number to watch in Phase 1, and the Phase 0 prototype should report measured endorsement latency against a live bank adapter rather than a synthetic peer. Standing computation adds no measurable load: it is deterministic derived state recomputed incrementally on each contributing event, touching only the affected entity's key.

Read and verification load. Verification never touches consensus. A foreign verification is a local BBS+ proof check (single-digit to low-tens of milliseconds for a handful of disclosed attributes), a revocation-status lookup against a CDN-cached bitstring, and a Merkle-path check of about sixteen hashes against a daily root of roughly 48,000 leaves. **End-to-end verification latency is well under 200 ms and scales with CDN capacity, not ledger capacity.** A review submission is a single nullifier existence check; a standing lookup is a single cached key read. Both are $O(1)$. **State growth** at roughly 500 bytes per anchor is about **9 GB per year**, or around 90 GB over a ten-year horizon; the nullifier registry adds 32 bytes per (worker, employer) pair, and a revocation bitstring covering 17 M credentials is about 2 MB served from cache. Public anchoring is one 32-byte transaction per day, paid in fiat — a fixed line item independent of volume.

The conclusion is structural, not merely arithmetic: the ledger's job is *integrity*, not throughput. Documents and payloads never touch it, verification is an off-chain proof check, and the only on-chain writes are small, batched anchors. This is why the design scales to national volume on a permissioned network with commodity infrastructure.

1. E. Androulaki et al., “Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains,” Proc. 13th EuroSys Conference (EuroSys '18), Porto, April 2018.
2. Nasrulin et al., “Gromit: Benchmarking the Performance and Scalability of Blockchain Systems,” arXiv:2208.11254.
3. P. Thakkar, S. Nathan, B. Viswanathan, “Performance Benchmarking and Optimizing Hyperledger Fabric Blockchain Platform,” Proc. IEEE MASCOTS 2018, pp. 264–276. DOI: 10.1109/MASCOTS.2018.00034.
4. Nguyen et al., 48-node commodity-cluster evaluation, as summarised in “Performance Tuning and Scaling Enterprise Blockchain Applications,” arXiv:1912.11456.
5. “Scalability and Efficiency Analysis of Hyperledger Fabric and Private Ethereum,” Computers 14(4):132, 2025.

7. Governance

Governance is organised against the three-part structure the domain requires: network membership, business network, and technology infrastructure.

7.1 Network membership governance

7.1.1 Member on- and off-boarding. Organisations apply with legal registration, sector licence, and a signed network charter. Admission requires a governance-channel vote plus **two-organisation endorsement** of the MSP identity issuance; agencies additionally require a valid AgencyLicence, and licence lapse suspends submission rights automatically. Off-boarding — voluntary exit, licence loss, or a misconduct finding — follows a fixed sequence: write suspension, a 90-day wind-down for in-flight credentials and applications, MSP revocation, and archival of endorsement and standing history. **Previously issued credentials remain valid and independently verifiable after an organisation leaves**, so a worker never loses proof because an institution closed. Standing history likewise survives exit, so a delisted agency cannot re-enter under a new name with a clean record while presenting the same beneficial ownership.

7.1.2 Regulatory oversight provisioning. The relevant regulator holds a **compliance veto** over admission in its own domain: BMET over labour-market issuers and agencies, RJSC over corporate issuers, BFIU over disclosure-consuming members. The veto is a negative power only — it can block an unfit member but cannot unilaterally admit one, and cannot alter or delete any anchored record. BMET, with the civil-society

observer, is also the endorsing pair for standing adjustments (§6.2), placing intermediary discipline with the body that already holds licensing authority rather than with a commercial party. Regulator nodes are anchored on the same terms as every other member.

The government trust paradox, addressed

§2.4 argues the platform must assume any official may be dishonest, yet §6.2 seats the government as a founding member with veto and standing-endorsement power. This is the design's central bet, and it rests on distinguishing **authority** from **custody**. Only BMET can clear a worker or licence an agency, only RJSC can attest legal existence, only DIFE can inspect — a system excluding the state would produce credentials no one is obliged to honour. What the state does *not* get is custody. **It cannot fabricate:** every non-sovereign schema needs a non-state co-signature (§6.2). **It cannot erase:** the ledger is append-only and the daily root is published beyond consortium control (§6.8), so the state's own record is anchored on the same terms as everyone else's. **It cannot surveil:** no plaintext, pairwise presentations, no bulk-export path — only scoped case evidence under legal authority or worker consent. **It cannot capture governance:** the veto is negative-only, and standing adjustment requires the independent observer's co-signature. We do not ask whether the government is trustworthy, because the architecture does not require an answer — and unlike an assurance, that guarantee survives a change of administration.

7.1.3 Permission structure. Four roles with distinct chaincode permissions. New member classes (RJSC, BFIU, auditors, foreign verifiers) map onto these rather than creating bespoke ones.

Role	May write	May not
Issuer	Credentials within its authorised schemas; applications where licensed	Read other issuers' PDC data; revoke another issuer's credential; adjust any standing
Verifier	Consent and disclosure logs; mismatch allegations	Write credentials; access vault plaintext; convert an allegation into a finding
Regulator	Sovereign credentials (clearance, registry, licence); standing adjustments with dual endorsement	Bulk-export worker records; alter history; adjust standing alone
Operator (foundation)	Schema registry and fee parameters, both under governance vote	Write, edit, or delete any credential, allegation, or score

7.1.4 Network operations. Each member runs its own peer and one orderer against a published availability target. Mandatory and scheduled: **quarterly key rotation**, an annual third-party security audit, incident-response runbooks with named contacts per organisation, geographically distributed peers across independent organisations, and disaster-recovery peers with documented RTO and RPO in the network SLA. No organisation holds a majority of orderers.

7.2 Business network governance

7.2.1 Business charter and operations structure. A **non-profit consortium foundation** holds the charter. Its board comprises the labour regulator, the corporate regulator, industry (BAIRA), banks, training institutions, a civil-society observer (ILO/IOM), and **a worker representative with a full vote**. No organisation holds a controlling share; charter amendment needs a supermajority plus regulator non-objection. The foundation is barred by charter from holding a commercial stake in any member, so its neutrality is structural rather than promised.

7.2.2 Common services management. The foundation operates and funds the shared layer: the schema registry (proposal, review, vote, versioning, deprecation), revocation infrastructure and status-list CDN, the reference wallet and verifier SDK, marketplace matching, standing computation and digest publication, the anchoring bridge, and dispute-support tooling. Each has a named owner, a published change process, and a deprecation policy with minimum notice. **The standing formula and its parameters are published, versioned, and changeable only by governance vote** — an agency can always know what is measured and how, which is a precondition for the score being fair as well as unbuyable. A schema is never mutated in place, only superseded, so a credential's meaning at issuance is permanently recoverable.

7.2.3 Business SLA and regulatory compliance. The published SLA covers verification availability and latency, issuance turnaround, **endorsing-peer response time per member** (the binding constraint in §6.10), **allegation response and adjudication windows**, status-list freshness, incident notification, and support escalation. **The Operations Suite ships under the same SLA and compliance schedule as any other network service** — a commercial product does not get a lighter regime. Obligations covered include labour law, AML/CFT reporting through BFIU, data protection (no PII on-chain, crypto-shredding erasure, pairwise DIDs), and cross-border transfer constraints for destination verifiers. Disputes — contested ServiceRecords and mismatch allegations under §3.8 — follow evidence collection, the subject's response window, mediation, then referral to the competent labour authority or court. An allegation withdrawn, unendorsed, or resolved in the agency's favour leaves no mark, and its resolution is itself anchored. The governing principle is a hard boundary: **the platform supplies evidence and never decides a case.**

7.3 Technology infrastructure governance

7.3.1 Distributed IT structure. Each organisation operates its own infrastructure under its own control; **government nodes remain physically inside government facilities**, a precondition for sovereign participation. The foundation runs only common services and holds no peer with privileged write access. Members are distributed across independent hosting environments so no single provider outage or jurisdiction can halt the network.

7.3.2 Technology assessment and adoption. Chaincode upgrades, Fabric version changes, cryptographic suite changes, standing-formula changes, and anchor-chain selection follow one pipeline: proposal, security review, testnet validation, member vote, staged rollout with rollback. **Crypto-agility is a governance requirement** — the signature suite is a schema field, credentials are re-issuable under new suites, and anchors are hash-algorithm-tagged, so a 40-year career horizon survives cryptographic obsolescence.

7.3.3 On-chain and off-chain data services. The §6.4 split is enforced by governance, not convention: **any proposed new on-chain field must pass a privacy review before deployment**, and the reviewing body includes the civil-society observer. Vault operations — envelope-encryption key custody, replication targets, retention schedules, and the crypto-shredding erasure path — are specified in the charter and audited annually. PDC membership lists, including those governing application detail, are themselves governance-controlled.

7.3.4 Risk mitigation. Scheduled backups with restore testing, annual third-party penetration testing, an annual network-wide risk assessment, and periodic incident-response exercises are mandatory. A **4-of-5 organisation agreement** is required to pause new writes in a major emergency — high enough that no faction can freeze the network, low enough to act on a live compromise. Reads and verification remain available during a write pause, so a worker at an airport is never stranded by a governance action.

7.4 Asset tokenisation model

The platform tokenises **four non-transferable asset classes**. There is no cryptocurrency and no tradable network token; all fees settle in fiat (BDT) through regulated rails.

Asset class	Unit	Transferable?	Economic purpose
Credential	One per claim, bound to subject DID	No	Prevents a market in credentials — the core fraud vector
Reputation stake	Issuer and agency standing	No — slashable, not sellable	Aligns issuers and intermediaries: good attestations build standing, false ones are slashed
Review-vote weight	One per (worker, employer), spent as a nullifier	No	One verified voice per employer — the anti-astroturfing unit
Logistics work-order	One per deployment or clearance task, lifecycle-stateful	No	Auditable unit of the Operations Suite; proves each step happened

Non-transferability is the security property, not a limitation: transferability would create a market in credentials, reputation, and votes — precisely the corruption the platform exists to remove. A sellable agency standing would reproduce, in digital form, exactly the licence-renting that makes the current recruitment market opaque. Anchoring fees are paid in fiat through a regulated provider (\$6.8), so no participant is exposed to a volatile digital asset.

8. Valuation and Distribution

8.1 Value created

Beneficiary	Value
Worker	Lower recruitment cost through disintermediation of unverifiable middlemen; a checkable answer to “is this agency honest?” before payment ; wage premium for provable skill; admissible evidence in wage and substitution disputes; access to formal credit on verified income; career history that survives every transition
Domestic employer / agency	Screening cost and time-to-hire collapse from weeks to seconds; applications that separate proven claims from asserted ones; a credible quality signal that lets compliant firms escape competition on price alone
Foreign employer; corporate customer (Operations Suite)	Confident hiring of verified skills with no local trust required and drastically lower mis-hire and fraud rates; a managed, compliant workforce and a verifiable reputation that wins government tenders and foreign contracts
Government	Fraud reduction, a higher-skill export mix, enforceable labour law, verifiable companies, live corridor analytics, an automated violation early-warning system, a graduated instrument for agency discipline below licence revocation , and a demonstrable worker-protection record in bilateral negotiations
Financial sector and national economy	A new creditworthy segment with verified income and measurably lower default risk on migrant lending; skill upgrading raises remittance per worker , the highest-leverage variable in the entire sector

Not all of this value is revenue, and deliberately so. The ministry early-warning dashboard, the public transparency digest, and worker-side access to agency standing are delivered as **public goods**, not revenue lines. Charging a worker to check whether an agency is honest would reproduce the extraction the platform exists to remove.

8.2 Revenue model and break-even

Stream	Payer	Basis	Indicative pricing
Corporate Operations Suite	Employers, agencies	Tiered monthly subscription by managed headcount, plus per-deployment logistics	BDT 150–200 per managed worker per month; BDT 1,000 per deployment — the primary stream
Verification-as-a-service	Employers, agencies, verifiers	Per verification, tiered by volume	BDT 100 per verification (see §4.1.1)
Marketplace screening	Employers	Per application screened, bundled at volume	Per-screening fee
Direct-hire success fee	Employers	Placements made without agency intermediation	Negotiated
Financial-services API	Banks, MFIs, insurers	Per income-verification call	BDT 25–50 per call
Regulatory licensing	Government	Per-clearance licence bundled into existing emigration clearance	BDT 50 per clearance
Aggregate analytics	Government, development agencies, researchers	Subscription to corridor and compliance intelligence	Subscription

Benchmarking against the domestic HRIS market. PiHR, the volume leader, prices at BDT 3,000 per month for 1–25 employees, BDT 5,000 for 26–50, and BDT 9,000 for 51–100; SmartHRM runs from BDT 2,000 per month for 25 members to BDT 4,500 per month for 300. Effective per-employee cost therefore ranges from roughly BDT 120 per month at the smallest tier down to BDT 15 per month at scale, and vendors offering explicit per-user pricing generally sit between BDT 50 and BDT 200 per user per month. Our BDT 150–200 sits at the **top of that band, and above it at scale** — a deliberate premium of roughly two to four times against a mid-market local HRIS.

The premium is not justified by features; a local vendor matches us on payroll and attendance. It is justified because the Suite produces an asset no HRIS can: an anchored, multi-party-endorsed record that converts into **procurement eligibility, foreign job orders, and subcontract awards**. For a recruiting agency, one additional job order won covers a year of subscription for several hundred managed workers; for an RMG supplier facing buyer compliance audits, a verifiable wage and safety record is audit cost avoided. We therefore expect the Suite to be unsellable to employers with no external counterparty to convince, and highly sellable to those bidding for government tenders or foreign contracts — the segment the go-to-market targets first.

A note on what is deliberately not monetised

An agency cannot pay for a higher standing, and no tier of the Suite confers one. This forgoes an obvious revenue line, and does so because the moment standing is purchasable it is worthless — to the worker checking it, to the employer relying on it, and therefore to the agency buying it. The refusal is the product.

Cost base and break-even. Node capital and operating costs are distributed — each member runs its own peer — so the foundation's cost base is engineering, common services and support, modelled at **BDT 50–60 M per year**. At BDT 100 per monetisable event, roughly **550,000 events per year** covers that — about **69,000 fully enrolled workers** at eight events each, reached inside Phase 2, *before* counting Suite subscriptions, marketplace fees, or API revenue. §4.1.1 gives the sensitivity across alternative fee levels.

The flywheel. A corporate customer pays for the Suite because it converts into contract value — verifiable reliability to foreign buyers and to government, therefore more tenders and job orders won, therefore more reason to pay. Separately, agencies drive credential issuance because attested claims carry no reputational risk while asserted ones do (§3.8), so the reputation mechanism generates the issuance volume the verification business needs. **The trust is earned from anchored evidence, never purchased.**

8.3 Go-to-market and roadmap

Phase	Window	Objective	Success measure
0 — Prototype	Months 0–6	Fabric network with three organisations; wallet and verifier SDK; one corridor, one trade; review prototype with working nullifier; marketplace field split and standing digest (read-only)	Working end-to-end issuance, ZK verification, review submission, and an application showing attested-versus-asserted separation; measured endorsement latency against a live bank adapter ; BCOLBD prototype round
1 — Corridor pilot	Months 6–18	Two licensed agencies, one TTC, one destination employer group; review surface beta; Operations Suite MVP; allegation and adjudication path live once the regulator node joins	5,000 workers enrolled; measured reduction in time-to-verify and screening cost; employer-side verification calls per enrolled worker ; ratio of attested to asserted fields per application
2 — Ops Suite GA + regulator	Months 18–30	BMET clearance credential live as standard output; agency licensing integrated; bank wage-event issuance; ministry trigger live	100,000+ workers; operational break-even crossed ; agency standing digests published network-wide
3 — Destination + corporate	Months 30–48	Bilateral recognition in one major corridor; RJSC and UBO registry; procurement transparency	Employer-side verification volume exceeds issuance volume — the network-effect inflection
4 — Regional	Year 4+	Cross-border W3C schema interoperability across South Asian sending states	Cross-border credential recognition; self-funding foundation

Sequencing note. The marketplace is deliberately split across two phases. Phase 0 ships only the parts that need no adjudicating authority — the attested and asserted field split and a read-only standing digest computed from existing events. The allegation and adjudication path waits for Phase 1, because StandingAdjustment requires a regulator endorser and there is no honest way to demonstrate that mechanism without one. Shipping the scoring half first would be a prototype of a system whose central safeguard does not yet exist.

Phase 0–1 funding requirement. Phases 0 and 1 span eighteen months and cost an estimated **BDT 65–80 M**: roughly BDT 45 M in engineering (a six-to-eight person team building chaincode, wallet, verifier SDK, marketplace, and the Suite MVP), BDT 15 M in pilot operations (enrolment logistics, TTC and agency onboarding, field support across the corridor), and BDT 10–20 M in security audit, legal, and consortium formation costs. Against this, Phase 0–1 revenue is approximately **BDT 5.5 M** — 5,000 enrolled workers generating on the order of BDT 4 M in verification events plus roughly BDT 1.5 M in early Suite subscriptions. **The funding gap is therefore BDT 60–75 M over eighteen months.**

We state this plainly because a business model claiming a pilot is self-funding is not one a jury should believe, and because the gap is small relative to the sector's existing programme spending. Target sources, in order of fit: **ILO and IOM** fair-recruitment programming, for which verifiable contract terms and a working intermediary-accountability mechanism are direct evidentiary outputs; the **World Bank and IFC** migration and remittance-cost portfolio, where the BDT 4.17 lakh average migration cost (§2.1) is an explicit target variable; **UNDP and bilateral development agencies** already funding digital-governance work with BMET; and **domestic bank co-investment** against the migrant-lending segment the wage credential unlocks.

If grant funding does not materialise, the fallback is to invert the sequence: launch the Operations Suite as a conventional commercial HRIS product with anchoring built in, sell it on workflow value alone, and fund consortium formation from revenue. This is slower and produces a weaker network in the interim — a single-issuer product in which the standing mechanism cannot yet operate for want of an endorser — but it is viable, because the Suite's workflow value stands independent of whether the consortium exists yet. The credential and reputation layers switch on when enough issuers and a regulator are live.

What would falsify this

The assumption that, if wrong, ends the project is that **destination employers will pay for verification rather than continuing to discount all claims** (§5.2). Phase 1 is designed to test exactly that: the pilot's primary metric is not enrolment but *employer-side verification calls per enrolled worker*. If that ratio stays near zero while credentials accumulate, the demand-side thesis has failed and the model must pivot to regulator-funded compliance infrastructure rather than verification revenue.

A second, narrower assumption is testable in the same window: that **workers will actually consult agency standing before choosing an intermediary**. If USSD standing lookups stay near zero while placements proceed, the accountability loop is open at the worker end, and the mechanism's force must come from employer-side screening and regulatory action rather than from worker choice. We would rather discover either of these in month 12 with 5,000 workers than in month 36 with 100,000.

9. Conclusion

Employment Passport began as a way to make a worker's credentials travel. It is now a corporate-transparency platform, because the same root cause — **institutions and paperwork that cannot be trusted** — corrupts every layer of Bangladesh's labour market at once: a worker's history, a company's identity, a company's reputation, the honesty of the intermediary between them, and the enforcement of the laws that protect workers.

A conventional system would have to pick a single administrator to trust, and in this environment that choice is exactly what fails. The platform therefore does something a database cannot: it spreads control across the institutions that distrust each other, signs every record with the parties responsible for it, makes every reputation a computed consequence of anchored evidence rather than an editable number, and anchors the whole history to a public chain so that even the platform cannot rewrite its own past.

The result is not a promise of honesty — it is a system in which honesty is no longer *required* for trust, because every claim can be verified against a proof that no single party controls, and in which honesty becomes the cheapest available strategy for the intermediaries who currently profit from its absence.

That is the difference between “trust us” and “verify the proof” — and it is the difference between a labour market that runs on forged paper and one that runs on provable facts.

Appendix A — Core chaincode interface

Function	Caller	Effect
RegisterDID (did, docHash)	Issuer (BMET)	Registers a DID document pointer on anchor-channel
IssueCredential (credHash, schemaId, issuerDID, subjectDID, expiry)	Authorised issuer	Anchors a credential; endorsement policy enforced per schema
VerifyAnchor (credHash)	Any verifier	Returns issuance state, issuer standing, endorsement set, corroboration score
SubmitReview (reviewHash, proof, nullifier, violationCode)	Worker wallet	Accepts a review only if the employment proof is valid and the nullifier is unused
SubmitApplication (subjectDID, orderRef, attestedRefs[], assertedHash, agencyDID?)	Worker or agency	Anchors an application; verifies attested references against the ledger; the agency signature binds it to the asserted-claim hash
FlagAssertionContradiction (applicationHash, credHash)	Chaincode (automatic)	Fires where an asserted claim is directly contradicted by an anchored credential; evidence is the contradiction itself
AllegeMismatch (applicationHash, evidenceHash)	Employer	Anchors an allegation only ; opens the 14-day agency response window; no standing effect
RespondToAllegation (allegationHash, counterClaimHash)	Agency	Anchors the agency's counter-claim within the response window
UpdateAgencyStanding (agencyDID, delta, evidenceHash)	Regulator + observer (2-of-2)	Adjusts reputation stake after the response window closes; dual endorsement required; rejects any call lacking both signatures
EmitViolationSignal (companyDID, code, evidenceHash)	Chaincode (threshold trigger)	Fires when a corroborated violation threshold is met; routes to the ministry
PublishDailyRoot (merkleRoot, thresholdSig)	4-of-6 founding organisations	Publishes the daily anchor; rejects any root without threshold signature
ComputeStandingDigest (entityDID) / ProveUBO (companyDID, thresholdProof)	Any member / company + auditor + RJSC	Recomputes standing deterministically from anchored events, so divergence is detectable; anchors a beneficial-ownership threshold proof
ProposeSchema / VoteSchema / SetFeeParameter / SetStandingParameter	Governance vote	Schema lifecycle; adjusts revenue-split, fee and standing-formula parameters. No unilateral change

Appendix B — Example credential schemas (abridged)

Fields marked **[D]** are disclosable independently under BBS+; fields marked **[O]** are held only off-chain.

EmploymentContractCredential

Field	Type	Notes
subjectDID	DID	[D] Worker identifier; pairwise per verifier
employerDID / agencyDID	DID	[D] Both required; both must endorse
jobTitle / tradeCode	string / ISCO-08	[D] Standardised occupational code for cross-border comparability
wageAmount / wageCurrency / wageBand	decimal / ISO 4217 / enum	[D] Band disclosable without the exact figure; supports ZK predicates
workingHoursPerWeek / overtimeTerms	integer / string	[D] Key evidence in substitution disputes
contractStart / contractTermMonths ; jurisdiction / governingLaw ; clearanceRef	date, integer / ISO 3166, string	[D] clearanceRef links to the BMET EmigrationClearance
contractDocument	encrypted blob ref	[O] Full signed PDF in the off-chain vault
workerPII	object	[O] Never on-chain in any form

ApplicationRecord

Field	Type	Notes
subjectDID	DID	[D] Pairwise, per employer
orderRef	hash	[D] The job order applied to
agencyDID	DID	[D] Present only for agency-mediated applications; absent for direct
attestedRefs[]	array of credHash	[D] Pointers to anchored credentials; auto-verified; no agency risk
assertedHash	hash	[D] Salted hash of the asserted-claim set; signed by the agency; this is what it stakes
agencySignature	signature	[D] Binds the agency to the asserted claims
assertedClaims	encrypted blob ref	[O] Full asserted-claim text; PDC-scoped to agency, employer, regulator
employerNotes	encrypted blob ref	[O] Internal screening notes; employer-only

WorkplaceReviewCredential

Field	Type	Notes
companyDID	DID	[D] The employer being reviewed
reviewHash	hash	[D] Salted hash of the review content
nullifier	scalar	[D] One-time PRF(linkSecret, companyDID); prevents duplicate voices
violationCode	enum	[D] Optional wage / contract / overtime / safety / termination flag
reviewText	encrypted blob ref	[O] Never on-chain; off-chain vault only

Appendix C — Glossary

BBS+ signature / DID / pairwise DID	BBS+ supports selective disclosure and zero-knowledge predicates over signed attributes. A DID is a decentralised identifier; <i>pairwise</i> means a distinct DID per verifier relationship, to prevent correlation.
Channel / private data collection / MSP	Fabric constructs that partition visibility across members. MSP is the Membership Service Provider, the certificate authority issuing network identities.
Nullifier / link secret	A nullifier is a one-time value PRF(linkSecret, companyDID) proving a unique voice without revealing identity. The link secret is a private per-employer value, committed inside the employment credential at issuance, which binds the nullifier and binds multiple credentials to one holder.
Attested versus asserted field	An attested field points to an anchored credential — automatically verified, carrying no risk for the submitting intermediary. An asserted field is backed only by the agency's signature, and is the unit on which agency standing is staked.
Allegation versus finding	An allegation is a claim anchored by one party and carrying no consequence; it becomes a finding only after the subject's response window and dual endorsement.
Standing	Deterministic derived state computed by chaincode from anchored events, combining volume, corroboration rate, upheld disputes, and counterparty diversity. Recomputable by any member; editable by none.
Merkle root / threshold signature	A Merkle root is a single hash committing to a set of records, published daily to a public chain. Its publication requires a threshold signature — valid only when at least 4 of 6 founding organisations jointly sign.
Endorsement policy / crypto-shredding / transparency digest	An endorsement policy is the rule defining which parties must co-sign a write for it to be valid. Crypto-shredding deletes an encryption key to render ciphertext permanently unreadable, satisfying erasure obligations without rewriting a ledger. A transparency digest is a public, Merkle-committed aggregate verifiable without raw data.

Appendix D — Data sources and references

- Migration and labour market.** BMET annual overseas employment statistics, 2025 (departures, destination distribution, skill-category breakdown), corroborated by *The Business Standard* and *Prothom Alo* reporting on BMET data, 2025–2026. Bangladesh Bureau of Statistics, *Survey on the Use of Remittances* (average migration cost, cost-recovery period). World Bank / KNOMAD, *Migration and Development Brief* (comparative worker-paid recruitment costs). IOM, *Bangladesh Household Remittance Survey* (cost distribution between licensed agencies and intermediaries). Deshi Mula, “About Us” (IT graduate supply versus demand).
- Destination-state verification.** *Prothom Alo*, “High migration cost mounts pressure on Saudi-bound migrants” (SVP fee; worker-borne intermediary costs of BDT 7,500–10,000). *The Business Standard*, “Dhaka seeks fee reduction for Takamol skills tests” (USD 50 test, USD 50 retake, formal request for reduction). *The Business Standard* / *Bonik Barta*, 2025 (mandatory from 19 June; five to thirty-three sectors; five-year validity). Migrant-Rights.org, “Saudi's Takamol makes millions on the back of migrant workers,” October 2024, and Business & Human Rights Resource Centre (TTC share of USD 13; 200,000+ accredited by October 2023; non-recognition of Bangladeshi certifications).
- Cost benchmarks.** Published Bangladesh attestation-chain pricing (BDT 3,000–4,000 per document). PiHR published pricing, mypihr.com/pricing (BDT 3,000/month for 1–25 employees; 5,000 for 26–50; 9,000 for 51–100). SmarHRM published pricing (BDT 2,000/month for 25 members to 4,500 for 300), with comparative Bangladesh HRIS surveys indicating BDT 50–200 per user per month for per-seat vendors.
- Hyperledger Fabric performance.** E. Androulaki et al., “Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains,” *Proc. 13th EuroSys Conference*, Porto, April 2018. Nasrulin et al., “Gromit: Benchmarking the Performance and Scalability of Blockchain Systems,” arXiv:2208.11254 (noting the EuroSys results used Zookeeper or PBFT ordering and are not directly comparable to Raft). P. Thakkar, S. Nathan, B. Viswanathan, “Performance Benchmarking and Optimizing Hyperledger Fabric Blockchain Platform,” *Proc. IEEE MASCOTS 2018*, pp. 264–276, DOI: 10.1109/MASCOTS.2018.00034. Nguyen et al., 48-node commodity-cluster evaluation, summarised in “Performance Tuning and Scaling Enterprise Blockchain Applications,” arXiv:1912.11456. “Scalability and Efficiency Analysis of Hyperledger Fabric and Private Ethereum,” *Computers* 14(4):132, 2025.